Name_____ Date_____ Period_____

LESSON 6-2 · HOMEWORK PACKET · SHEET 1 OF 2

Dividing a Fraction by a Fraction

Maryland Standard 6.NS.A.1 · Learning target – I can divide a fraction by another fraction using the reciprocal.

REMEMBER

- To divide by a fraction, **multiply by its reciprocal**: 3/4 ÷ 3/8 = 3/4 × 8/3.
- Keep the first fraction, change ÷ to ×, and flip only the **second** fraction.
- Check: quotient × divisor = dividend. 2 × 3/8 = 6/8 = 3/4.

WORKED EXAMPLE

How many 3/8-cup servings are in 3/4 cup of yogurt?

Step 1 Write the division: 3/4 ÷ 3/8

Step 2 Multiply by the reciprocal: 3/4 × 8/3 = 24/12

Step 3 Simplify: 24/12 = 2

Answer **2 servings**

PRACTICE

show your work · label units

1. The bar is 1 whole. The shaded part is 4/5 of the bar. How many 1/10 pieces fit in 4/5? Write the division and the quotient.



1 whole

Division_____

2. Divide. Write each quotient in simplest form.

a. 3/4 ÷ 1/8 = _____ **b.** 2/3 ÷ 2/9 = _____ **c.** 5/6 ÷ 1/3 = _____

d. 7/10 ÷ 2/5 = _____ **e.** 1/2 ÷ 3/4 = _____ **f.** 8/9 ÷ 4/9 = _____

3. **Pancakes.** A pancake recipe needs 2/3 cup of milk. Rosa has only 1/2 cup of milk. What fraction of the recipe can she make?

Show your work.

Answer_____ of the recipe

4. **Bookmarks.** Jae has 5/6 yard of ribbon. Each bookmark needs 5/12 yard. How many bookmarks can Jae make?

Answer_____ bookmarks

5. **Find the Error.** Imani wrote 4/5 ÷ 2/3 = 4/5 × 2/3 = 8/15. Explain her mistake. Then find the correct quotient.

_____

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME With a measuring cup, show someone at home how many 1/4-cup scoops are in 3/4 cup.

Name_____ Date_____ Period_____

LESSON 6-2 · HOMEWORK PACKET · SHEET 2 OF 2

Dividing with Mixed Numbers

Maryland Standard 6.NS.A.1 · Learning target – I can divide mixed numbers by writing them as improper fractions first.

REMEMBER

- Write each mixed number as an **improper fraction**: 2 3/4 = 11/4.
- Then multiply by the **reciprocal** of the divisor.
- Simplify. Write an improper answer as a mixed number.

WORKED EXAMPLE

A board is 5 1/4 feet long. How many 3/4-foot pieces can be cut?

Step 1 5 1/4 = 21/4, so 21/4 ÷ 3/4 = 21/4 × 4/3

Step 2 Multiply and simplify: 84/12 = 7

Answer **7 pieces**

PRACTICE

show your work · label units

1. Divide. Write each quotient in simplest form.

a. 4 1/2 ÷ 3/4 = ____ **b.** 3 1/3 ÷ 5/6 = ____ **c.** 2 1/4 ÷ 1 1/2 = ____

d. 6 ÷ 1 1/3 = ____ **e.** 5 5/6 ÷ 1 1/6 = ____ **f.** 1 7/8 ÷ 2 1/2 = ____

2. Write each mixed number as an improper fraction.

a. 3 2/5 = _____ **b.** 4 5/6 = _____

3. **Soup Bowls.** A pot holds 8 3/4 cups of soup. Each bowl holds 1 1/2 cups. How many bowls can be filled completely? How much soup is left over?

Show your work.

Answer_____

4. **Hiking.** A trail is 7 1/2 miles long. Kenji hikes 1 1/4 miles each hour. How many hours will it take him to hike the whole trail?

Answer_____ hours

5. **Reason.** Without dividing, decide: is 6 1/2 ÷ 1 1/4 greater than or less than 6 1/2?

☐ greater than 6 1/2 ☐ less than 6 1/2

TEST PRACTICE

choose the best answer

6. What is 3 1/2 ÷ 1 3/4?

A 1/2 **B** 2 **C** 3 2/3 **D** 6 1/8

7. A rope is 6 2/3 feet long. It is cut into pieces that are each 5/6 foot long. How many pieces are there?

A 1/8 **B** 5 5/9 **C** 8 **D** 12

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Measure 1 1/2 cups of water with a 1/4-cup scoop. Does the number of scoops match 1 1/2 ÷ 1/4?